

Predicting Student Success: A Data Driven Approach to Improving Retention

An advanced analytics model to proactively identify at-risk students and enable targeted, timely interventions.

The High Cost of Attrition: An Institutional Challenge

The Business Problem

- How can an institution optimise its student support services to improve retention rates and protect tuition revenue?
- Faculty and counsellors have limited time; they require a prioritized list of students who are most likely to drop out, allowing for proactive intervention.

Key Performance Indicators (KPIs) for Success



Retention Uplift

The measurable percentage increase in students who remain enrolled year-over-year compared to a pre-AI baseline.



Intervention Lead Time

The number of weeks of advance warning the model provides before final exams, giving counsellors a viable window to act. A minimum 4-week window is the goal.



Cost-Benefit Ratio

The financial gain from retained tuition versus the cost of intervention and model maintenance.

Framing the Data Science Solution

The Learning Task

- Type: Supervised Binary Classification.
- Objective: To predict whether a student will ultimately 'Dropout' or 'Graduate'.
- Target Variable: Binary outcome derived from the student's final status.

The Data Foundation

Source: The dataset originates from a higher education institution in Portugal, encompassing a rich set of multimodal features.

Scope: It contains 4,424 records of undergraduate students across 17 degree programmes, with 37 distinct features including demographics, prior academic qualifications, and first-year performance data.

Defining Technical Success

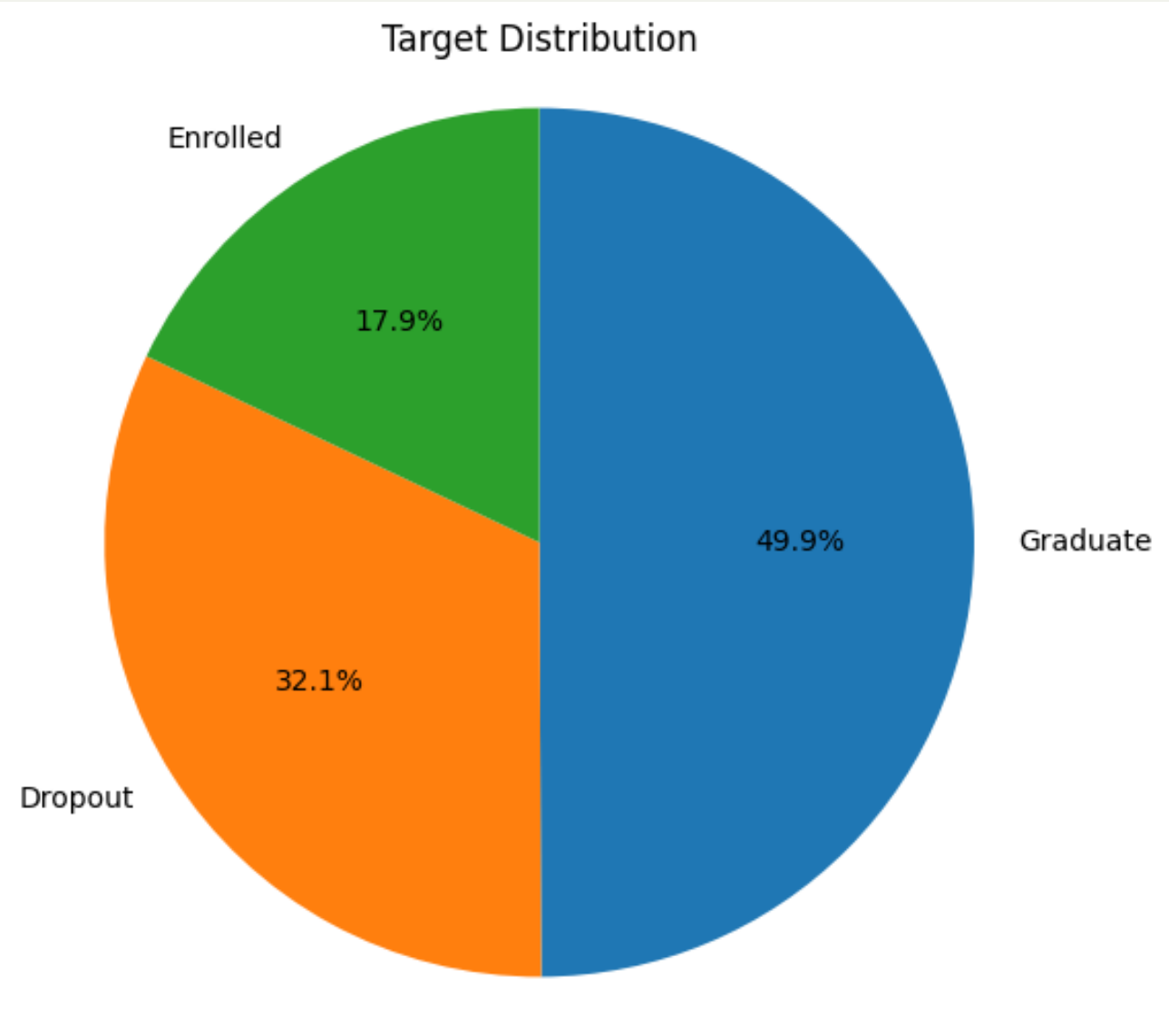
Given the high cost of missing a student who is at risk (a False Negative), the model will be optimised to maximise Recall and Area Under the Precision-RecallCurve (AUPRC), rather than simple accuracy.

Recall (Sensitivity) is the primary technical metric, as the goal is to identify the highest possible percentage of true at-risk students.

Understanding the Target Population

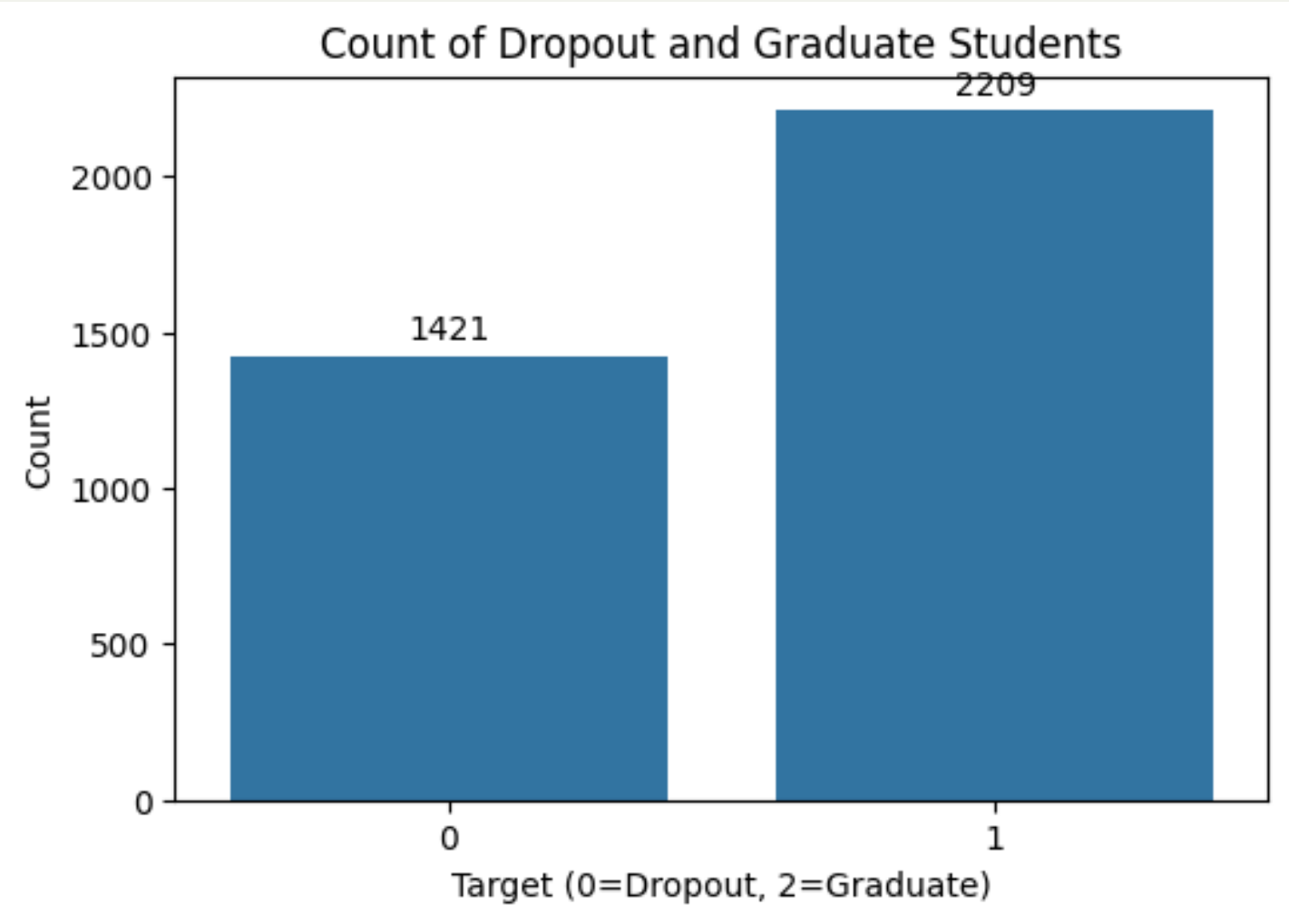
The raw dataset includes three student outcomes: 'Graduate', 'Dropout', and 'Enrolled'. For a predictive model, we focus on the two definitive final outcomes.

Initial Distribution of All Students



Class Imbalance

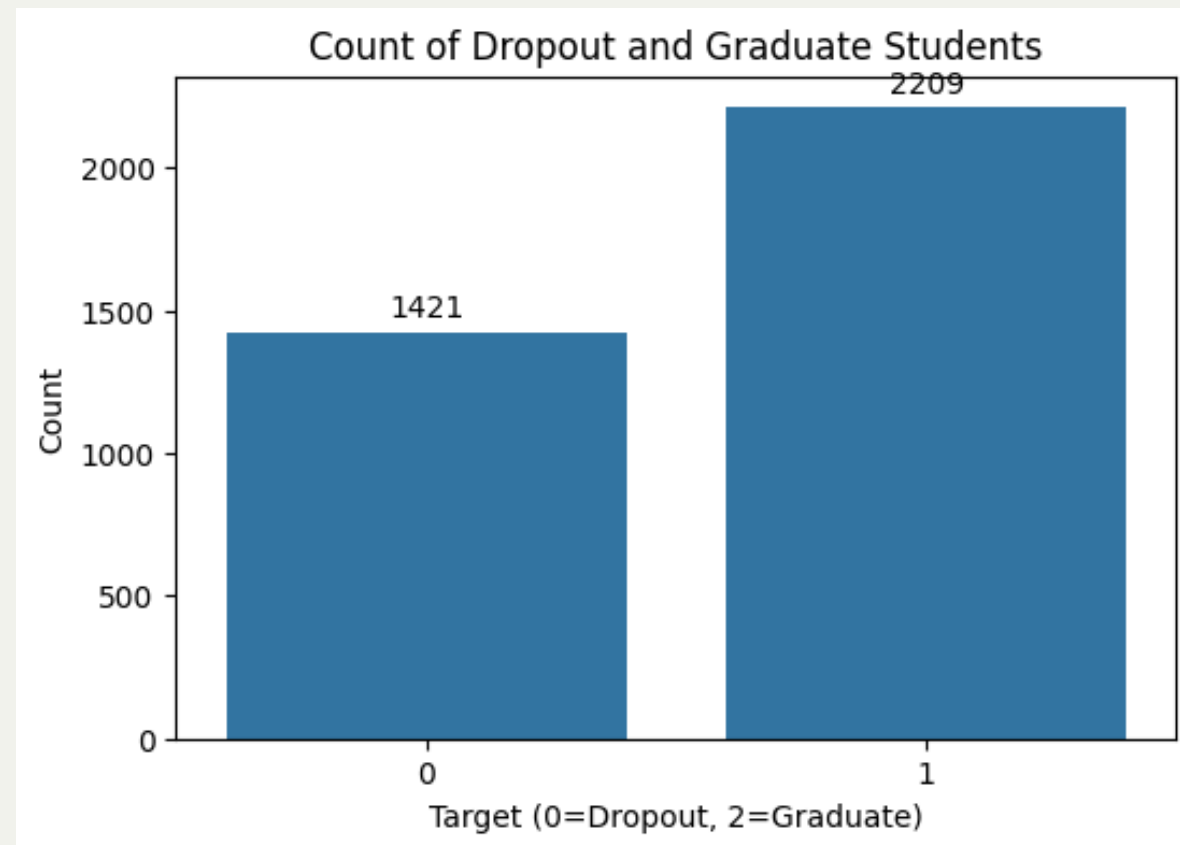
After filtering for definitive outcomes, a significant class imbalance is evident. The 'Graduate' class is considerably larger than the 'Dropout' class.



Creating a Balanced Foundation for Modelling

Training a model on an imbalanced dataset would create a bias towards the majority class ('Graduate'), severely compromising its ability to detect the minority class ('Dropout'), which is our primary objective. To solve this, the minority class was oversampled to match the majority class, creating a perfectly balanced dataset for training.

Count of Dropout and Graduate Students

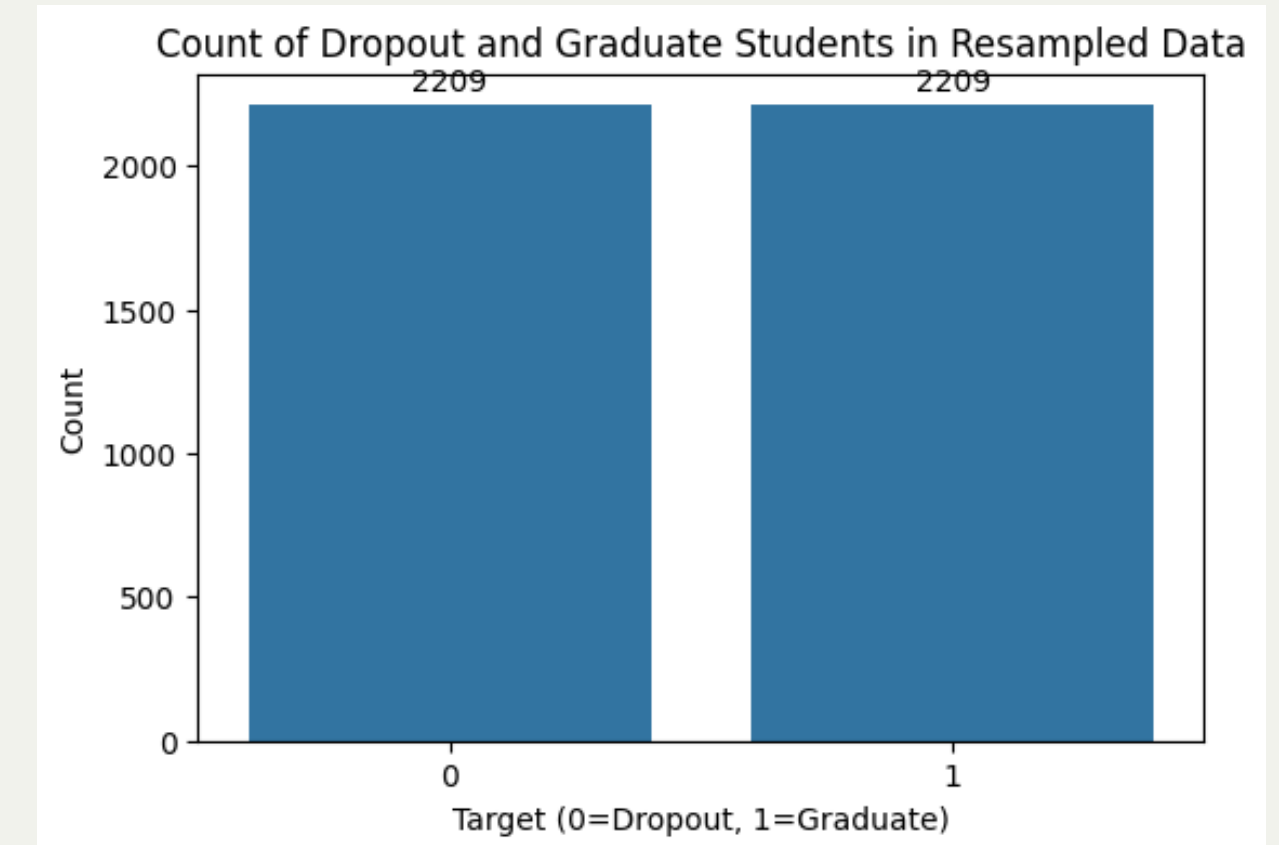


Original Imbalanced Data

Before vs. After



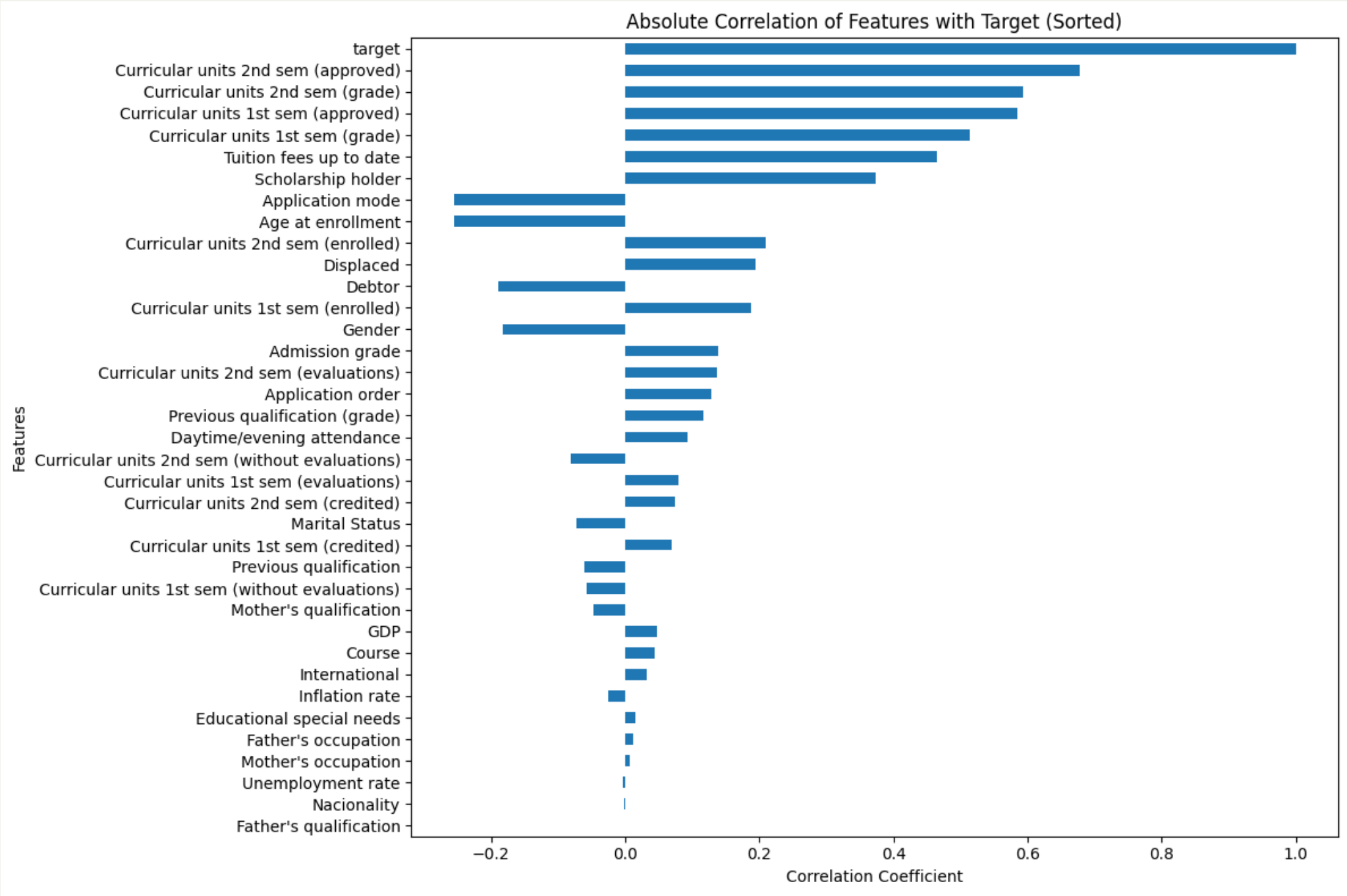
Count of Dropout and Graduate Students in Resampled Data



Resampled Balanced Data for Training

What Factors Drive Student Success? A Look at Feature Correlation

An analysis of the absolute correlation between each feature and the target outcome reveals the key predictors of student success. Academic performance indicators from the first and second semesters are overwhelmingly the most influential factors.



Top Predictors

The strongest signals come from academic achievement: "Curricular units 2nd sem (approved)", "Curricular units 2nd sem (grade)", and "Curricular units 1st sem (approved)".

Financial Signal

"Tuition fees up to date" also shows a strong positive correlation with graduating.

Demographic Signal

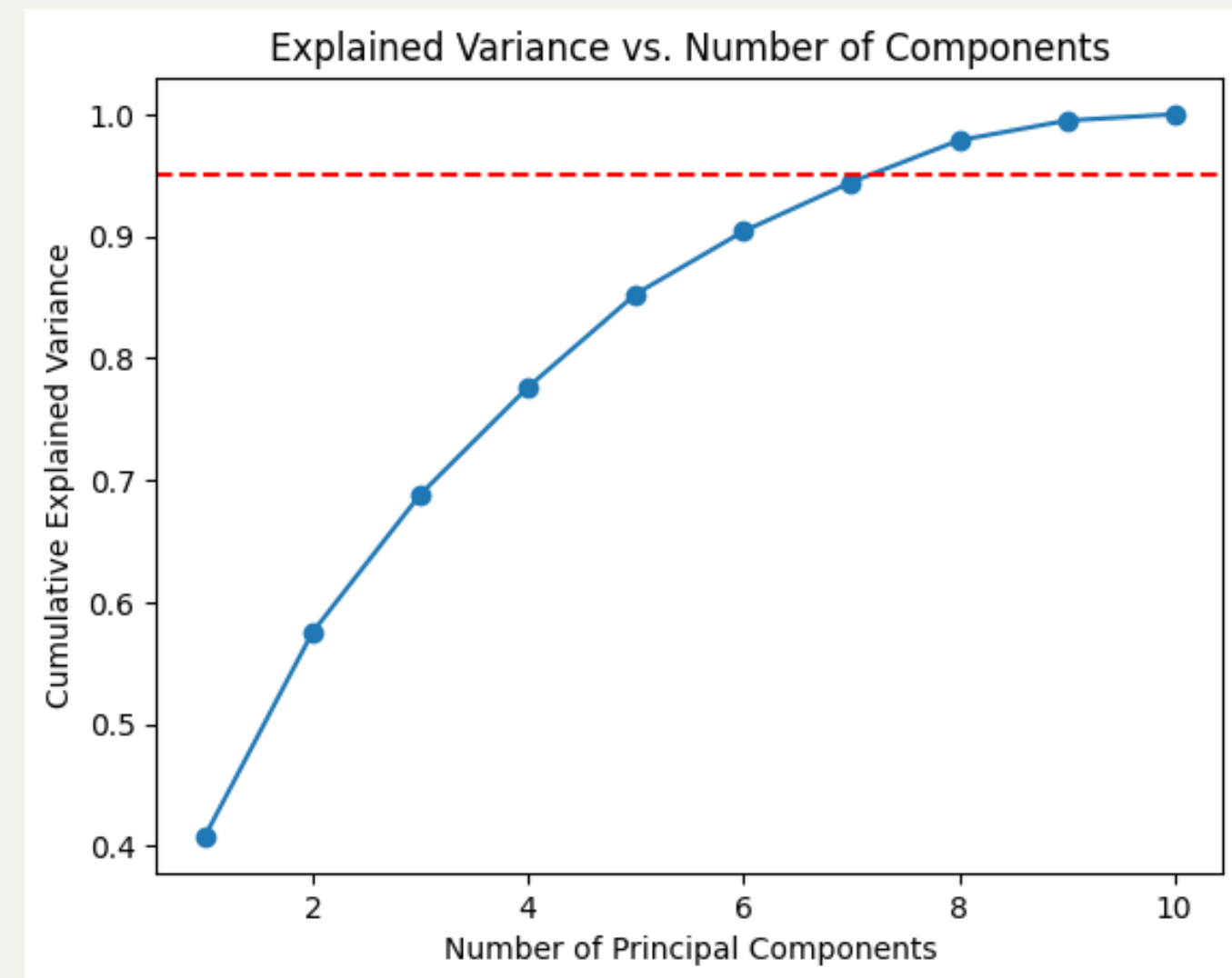
"Age at enrollment" is a notable factor, with older students showing a higher tendency to drop out.

Feature Engineering

With the use of Recursive Feature Elimination (RFE) , we have selected the top 10 most impactful features namely:

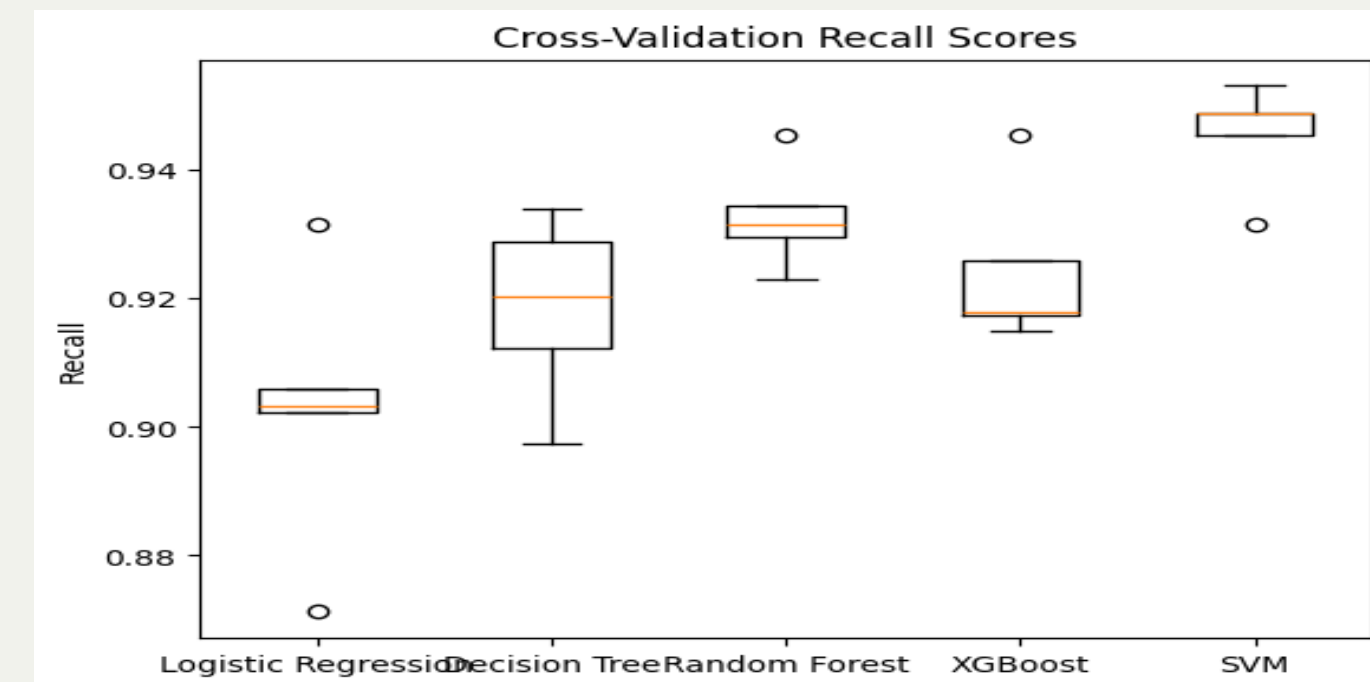
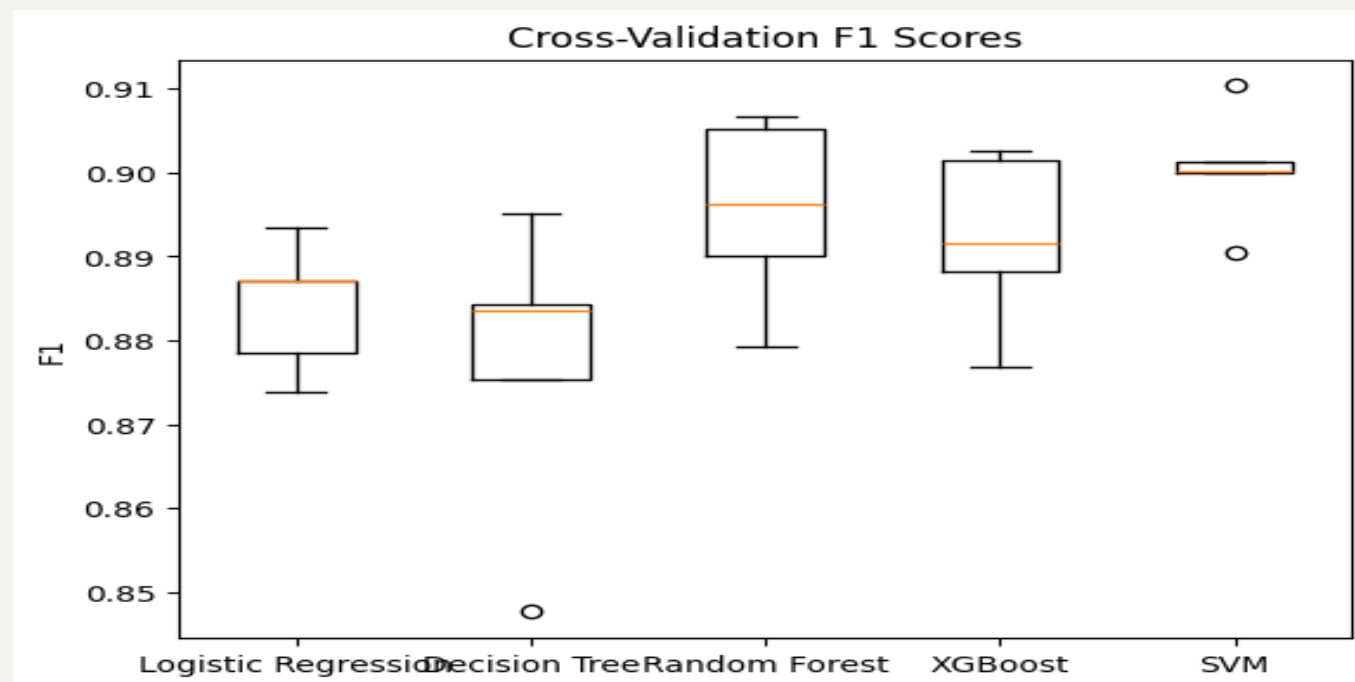
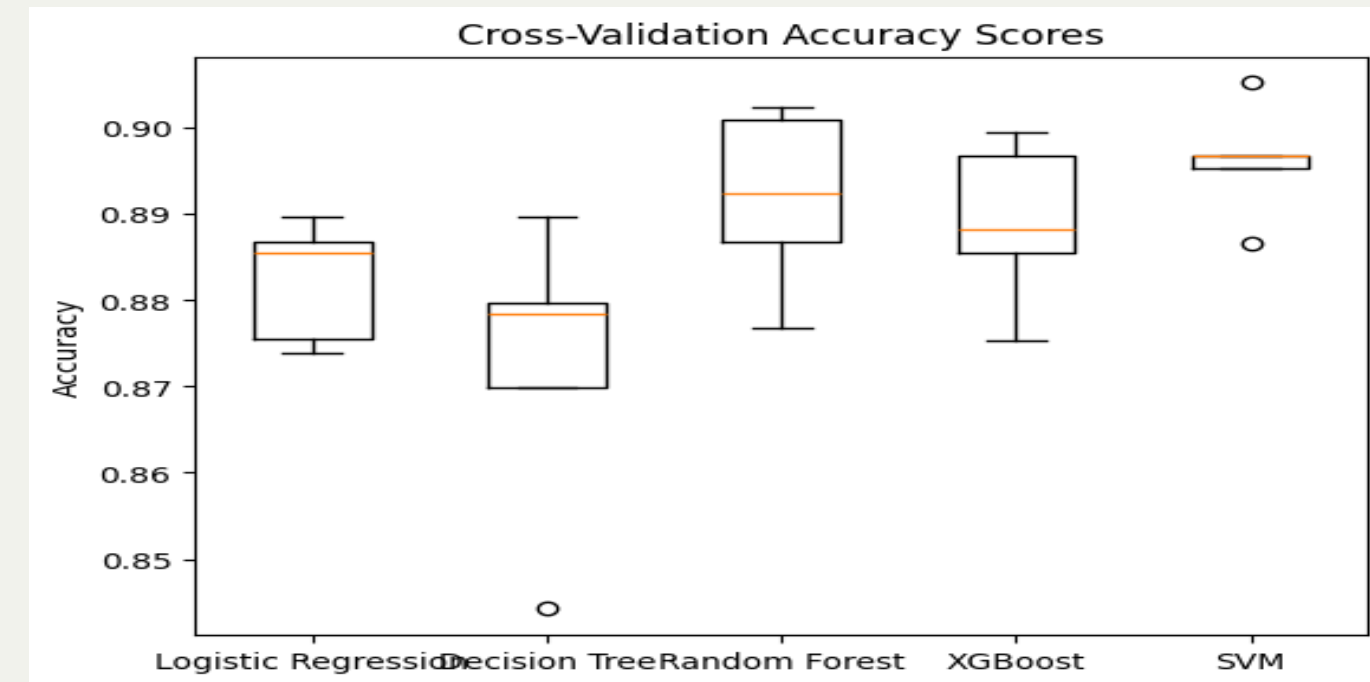
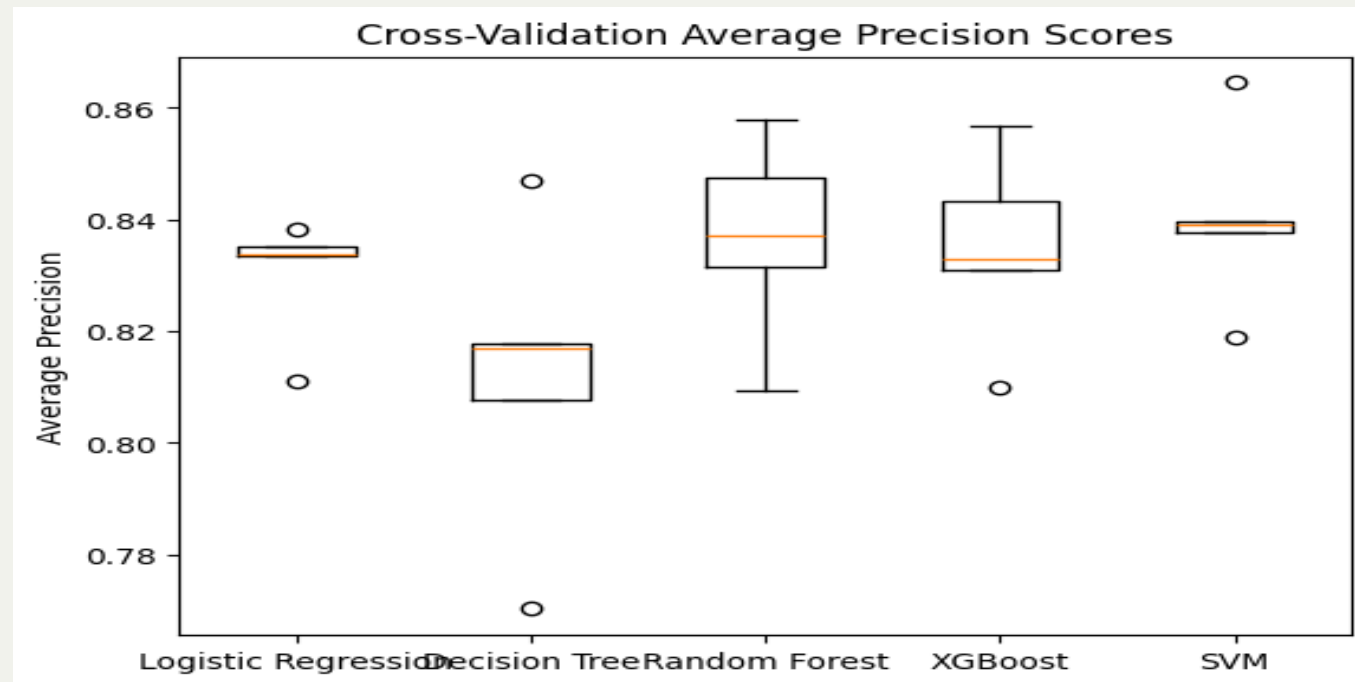
1. Course
2. Previous qualification (grade)
3. Admission grade
4. Tuition fees up to date
5. Age at enrollment
6. Curricular units 1st sem (approved)
7. Curricular units 1st sem (grade)
8. Curricular units 2nd sem (evaluations)
9. Curricular units 2nd sem (approved)
10. Curricular units 2nd sem (grade)

To further reduced the number of features we decided to implement Principal Component Analysis (PCA). We have chosen to reduce the features to 8 PCA components because it is the least number of components that will provide more than 95% cumulative variance.



A Tournament of Champions: Selecting the Best Predictive Model

Five distinct classification models were trained (with the best Hyper-Parameters) and evaluated using 5-fold cross-validation to ensure robust and generalizable performance. The models were compared on four key metrics critical for this classification task.



The **SVM** model demonstrates consistently high performance across all key metrics, particularly showing a highest median and tight interquartile range for Recall and F1-Score, indicating both strong predictive power and stability.

The Chosen Solution: A Fine-Tuned Random Forest Model

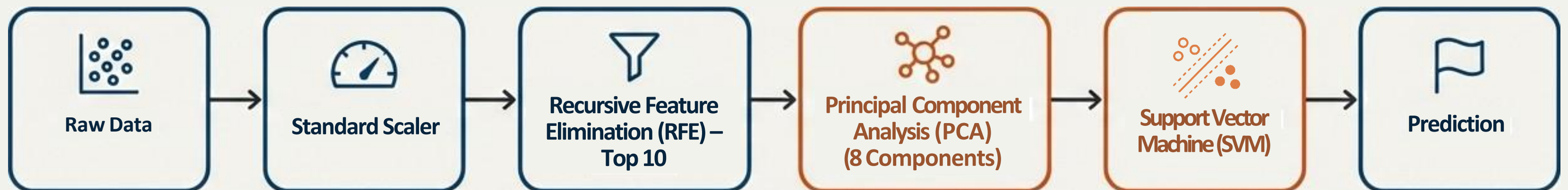
Based on its superior and highly stable performance during cross-validation, the Support Vector Machine (SVM) classifier was selected as the final model. It was then trained on the full pre-processed dataset.

Winning Model Performance

The model achieved an outstanding Recall of **0.95** and an F1-Score of **0.91** for the 'Graduate' class on the held-out test set, demonstrating its effectiveness at correctly identifying students on the path to success. Crucially, for the 'Dropout' class, it maintained high performance, ensuring that at-risk students are not missed.

The Final Engineered Pipeline (svc_pipeline.joblib)

The entire workflow, from data preparation to the final prediction, has been encapsulated into a single, serialised pipeline object for robust deployment.



Building Trust: Understanding the Model's Logic on a Global Scale

To ensure the model is not a 'black box', **SHAP** (**SH**apley **A**dditive **eX**planations) values were calculated. This reveals how each feature impacts the model's output across the entire dataset. The plot below shows the top 20 features, ranked by their importance.

Interpretation

- **Horizontal Axis**

The SHAP value (impact on model output). Positive values push the prediction towards "Dropout".

- **Feature Importance**

"Curricular units 2nd sem (approved)" is the most important feature.

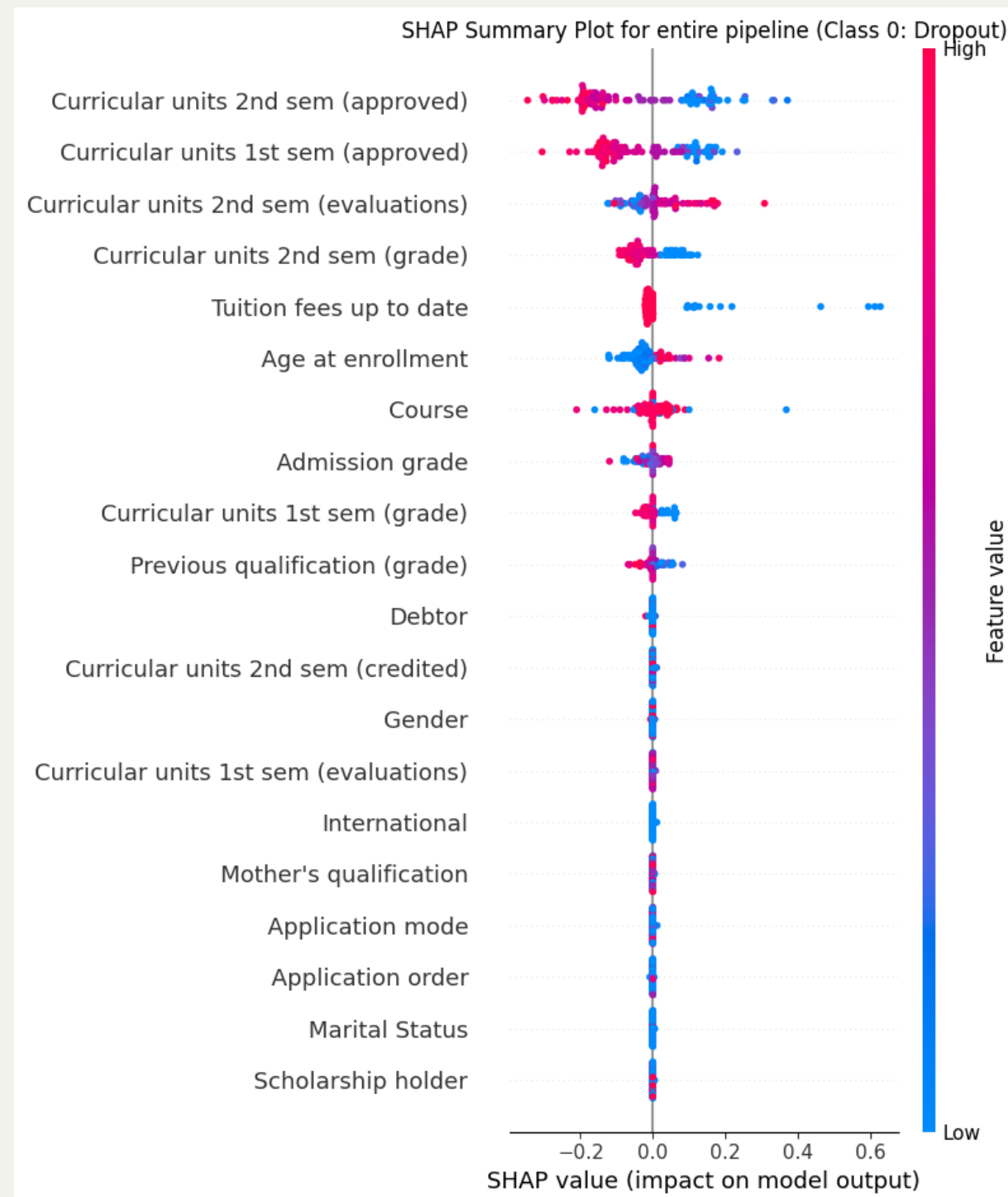
- **How to Read**

- Y-axis: Features ranked by importance (top = most influential)
- X-axis: SHAP value = impact on model output (positive = pushes toward dropout, negative = pushes away)
- Dots: Each dot = one student
 - Color: Feature value (blue = low, red = high)
 - Position: SHAP value (left = pushes toward dropout, right = pushes away)

For the top feature, low values (blue dots) have a high positive SHAP value, strongly increasing the likelihood of a "Dropout" prediction. High values (red dots) have a negative SHAP value, decreasing that likelihood.

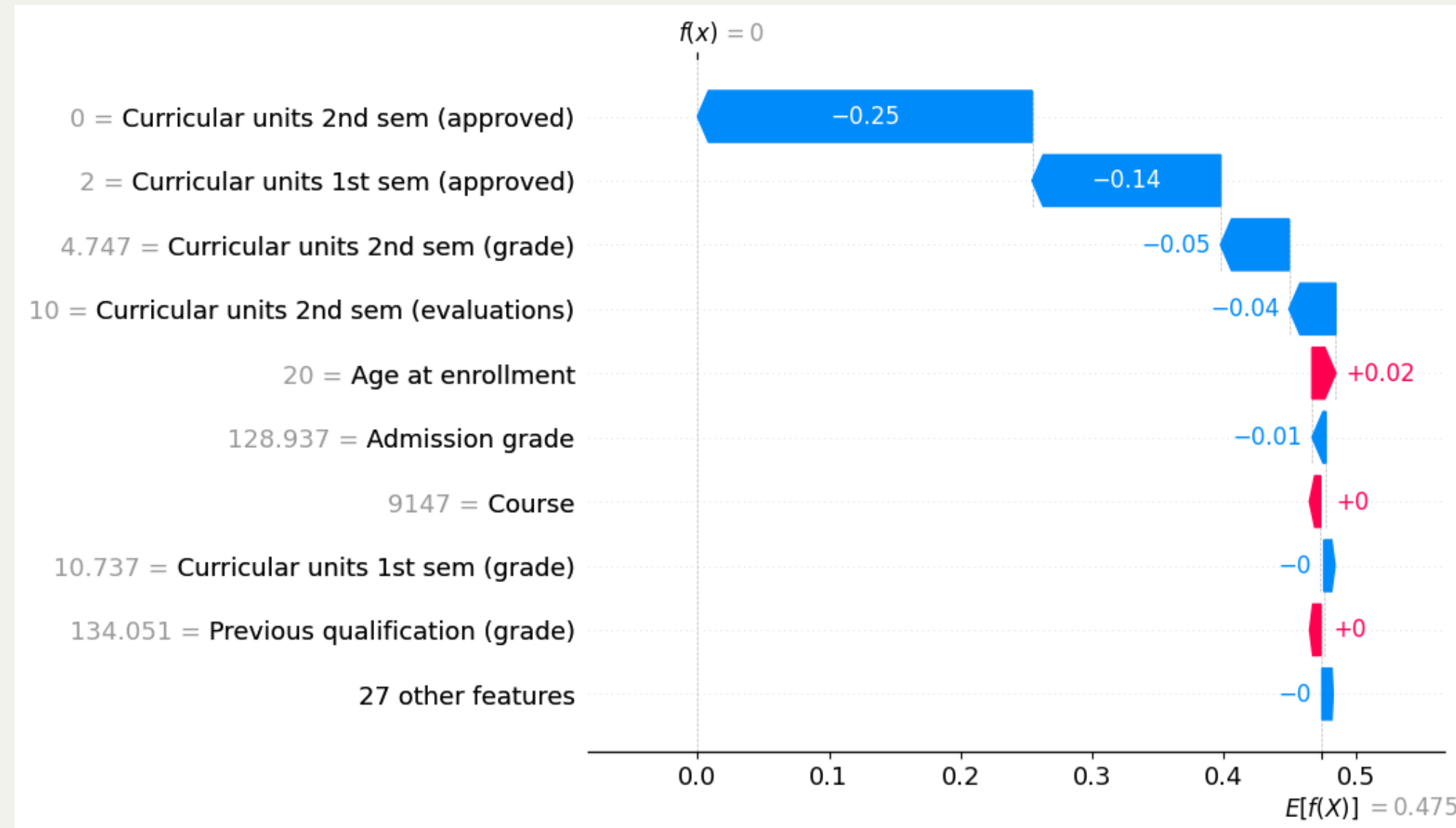
- **Key Insight**

Poor academic performance (low approved units, low grades) is the single most powerful driver of a 'Dropout' prediction.



From Global Trends to an Individual Story: Explaining a Single Prediction

SHAP can also explain the prediction for a single student. The waterfall plot below shows the features that pushed the model's prediction away from the baseline (average prediction) to its final decision for one specific student flagged as being at risk of dropping out.



Case Study Analysis

- **Baseline**

The average prediction probability is 0.475 ($E[f(X)]$).

- **Negative Drivers (Pushing towards Dropout)**

This student's outcome was heavily influenced by having 0 approved units in the 2nd semester (-0.25 impact) and 0 approved units in the 1st semester (-0.14 impact). These factors were the primary reasons for the at-risk flag.

- **Positive Drivers (Pushing towards Graduate)**

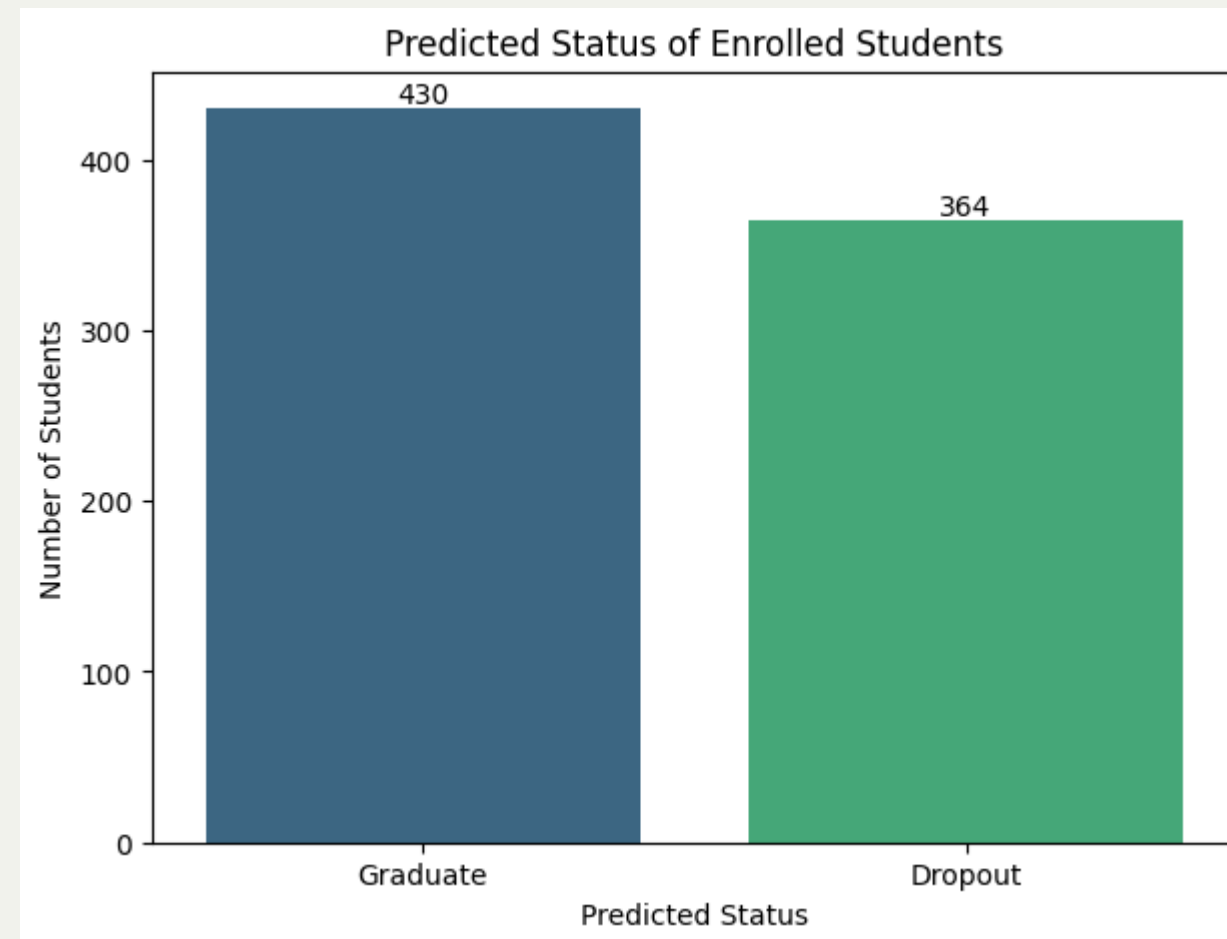
A relatively high 'Age at enrollment' (+0.02 impact) slightly counteracted the negative factors but was not strong enough to change the outcome.

- **Final Prediction**

The sum of these feature impacts results in the final prediction of $f(x) = 0$ ("Dropout").

The Payoff: Predicting the Future for Currently Enrolled Students

The true value of the model lies in its application to students whose outcomes are not yet known. The final pipeline was used to predict the future status of the 794 students currently marked as “Enrolled”.



“Predicted to Graduate”: 430 students are predicted to be on track for graduation.

“Predicted to Dropout”: 364 students are flagged by the model as being at high risk of attrition.

Key Takeaway: The model has generated a high-priority, data-validated list of 364 students for immediate proactive outreach by academic advisors and support staff.

From Prediction to Proactive Support: Summary and Future Directions

Project Impact Summary

Problem:

High student attrition presents a significant institutional and financial challenge.

Solution:

A robust and interpretable SVM model, encapsulated in a deployment-ready pipeline, was developed to predict student outcomes.

Impact:

The model provides an actionable list of **364 currently enrolled students** at high risk of attrition, enabling a shift from reactive to proactive support and optimising the use of counselling resources.

Recommendations and Next Steps



Deployment:

Integrate the 'svc_pipeline.joblib' model into a secure dashboard for academic advisors to view at-risk student lists in near real-time.



Refinement:

Continuously retrain the model with new student data each semester to adapt to changing trends and maintain high accuracy.



Expansion:

Explore the integration of additional data sources, such as virtual learning environment (VLE) engagement logs, to further enrich the feature space and enhance predictive power.